

FRONTEND DESIGN SPECIFICATION

For Figma — Interactive Mobile Frontend

SME Cash-Flow Prediction & Action System

This document is the frontend brief and buildable specification for the Figma prototype: design principles, information architecture, screen-by-screen wireframes with rationale, the design system (tokens, components), interaction and state rules, and a concrete plan for organizing the Figma file itself — pages, components, variants, and a clickable prototype path for a demo. Every screen decision is traced back to a specific technical behavior of the prediction/action engine (Sections 4-6 of the Technical Design Document), so the interface never claims more certainty than the model actually has.

1. Design Principles (derived from our specific constraints)

A generic fintech dashboard template will actively work against this product. Four constraints from the prediction system directly shape frontend decisions:

- **1. Never show false precision.** Every number the model produces is a probability, not a fact (see Technical Design Doc, Section 4-5). The UI must always pair a number with its confidence — a probability band on charts, a percentage on risk chips, a visible confidence_stage badge — never a bare number presented as certain.
- **2. Earn trust progressively, and show that visibly.** A brand-new business starts in Cold Start mode (cohort prior only) and graduates to Personalized as real invoices accumulate. The interface must make this state visible and honest at every screen, not hide it.
- **3. Prediction without action is worthless — so action must never be more than one tap away from a prediction.** Every risk/shortfall surface links directly to a ranked action list, and every action links directly to a pre-drafted, ready-to-send message. No screen should ever just say 'this is risky' and stop there.
- **4. Design for a phone, one thumb, possibly patchy connectivity, and a non-technical owner.** The target user (Problem Statement, Section 1) is a shopkeeper or small manufacturer, not a CFO. Large touch targets, minimal jargon, WhatsApp-native interaction patterns (since that's already how this audience communicates with customers), and graceful behavior when a screen loads slowly or offline.

2. Information Architecture

Four-tab bottom navigation, chosen over a hamburger menu specifically because the target user is mobile-first and unfamiliar with buried navigation — everything important should be reachable in one tap from anywhere.

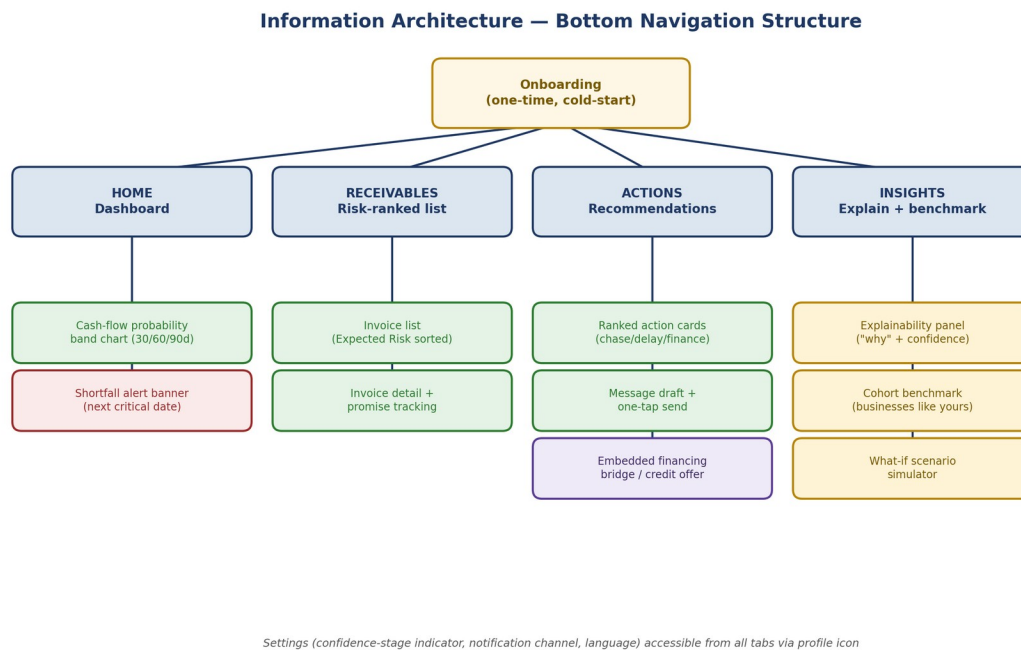


Figure 1. Site map / information architecture. Onboarding runs once before the four-tab structure takes over.

2.1 Why these four tabs, specifically

- **Home** answers 'am I okay right now' in one glance — the probability band chart plus the single most urgent alert.
- **Receivables** answers 'who owes me and how worried should I be about each one' — the Expected-Risk ranked list is the core object of the whole product.
- **Actions** answers 'what do I do about it' — this is where the optimization engine's output lives, and where a recommendation becomes a sent message.
- **Insights** is where trust-building content lives — explainability, cohort benchmarking, and the what-if simulator — deliberately separated from the urgent Home/Actions path so it doesn't clutter the moment of decision-making, but is always available.

3. Screen-by-Screen Specification

Seven core screens cover the entire critical path end to end: onboarding through cold start, the daily glance, a detected shortfall, taking action, understanding why, and handling a broken promise. Each is shown as a low-fidelity wireframe with annotations — build these first in Figma before any visual polish.

3.1 Screens 1-4: The Core Loop

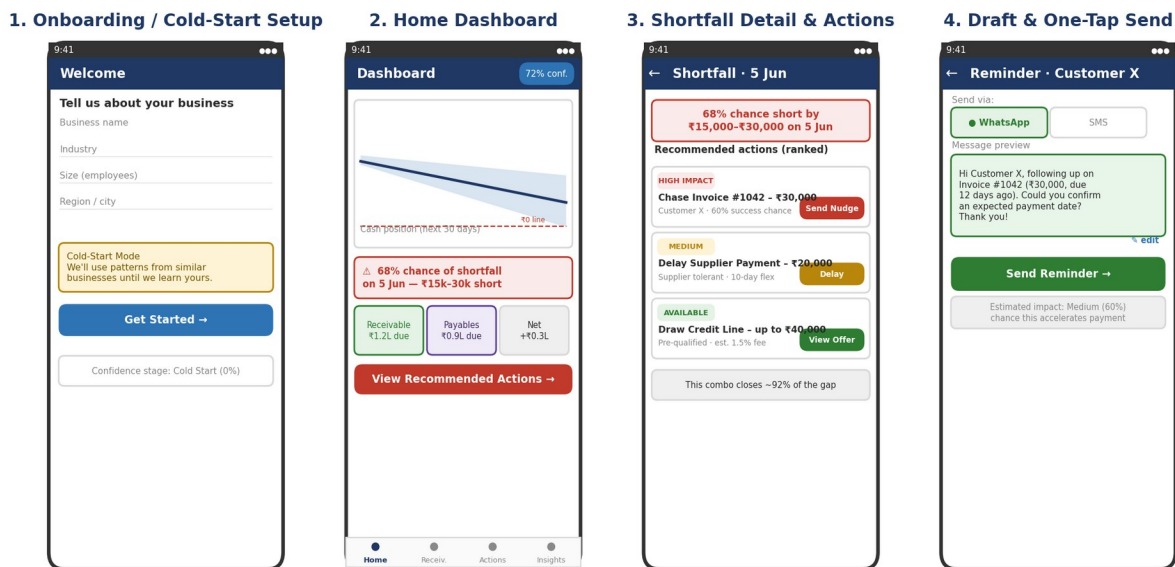


Figure 2. Screens 1-4 — Onboarding, Home Dashboard, Shortfall Detail & Actions, Message Draft & Send.

Screen 1 — Onboarding / Cold-Start Setup

- **Purpose:** collect the minimum fields needed to assign a cohort (industry, size, region) so the system is useful from message one.
- **Key element:** the amber 'Cold-Start Mode' callout, set immediately — this is the first moment the product tells the truth about its own confidence, and it should read as reassuring, not as a disclaimer buried in fine print.
- **State to design:** a returning-user variant of this screen never appears again after first setup; only the confidence badge in the app bar persists as a permanent reminder of confidence stage.

Screen 2 — Home Dashboard

- **Purpose:** the daily 5-second glance. Answers 'am I okay' before anything else.
- **Key element:** the probability band chart (shaded region, not a single line) is the single most important visual motif in the whole product — it should be reused everywhere a forecast appears, so the user learns to read it once and recognize it everywhere.
- **Key element:** the confidence chip top-right ('72% conf.') is small but permanent — never let a user forget which confidence stage they're in.
- **States to design:** no-risk-detected (green, calm, no red banner), single-risk-detected (as shown), multiple-risks-detected (banner becomes a small stack/carousel), and cold-start (chart shows a visibly wider band with a label explaining why).

Screen 3 — Shortfall Detail & Recommended Actions

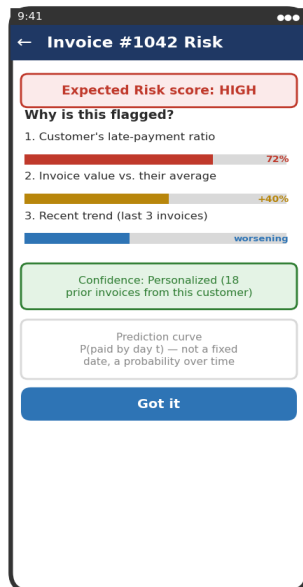
- **Purpose:** this is the money screen — where the optimization engine's output (Technical Design Doc, Section 6.2) becomes human-readable.
- **Key element:** each action card shows an impact tier chip (HIGH/MEDIUM/AVAILABLE) using the semantic risk palette, a one-line rationale (never just 'chase this customer' — always 'why this one'), and a single clear CTA button.
- **Key element:** the bottom summary strip ('this combo closes ~92% of the gap') closes the loop — the user should never wonder whether following the suggestions is actually enough.
- **State to design:** what happens when no combination of available actions fully closes the gap — this should visually route toward the Embedded Financing Bridge card rather than leaving the user with an unresolved number.

Screen 4 — Message Draft & One-Tap Send

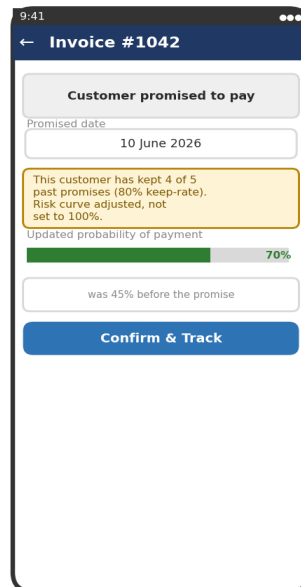
- **Purpose:** convert a recommendation into a sent message with the least possible friction — this is the single highest-leverage screen for a live demo.
- **Key element:** channel toggle (WhatsApp / SMS) defaults to whichever channel that specific customer has historically responded to fastest, once that data exists — personalization applied to delivery, not just prediction.
- **Key element:** the message tone is deliberately non-confrontational by template design — this is a direct interface expression of the relationship_cost term in the optimizer's objective function.
- **Key element:** 'Estimated impact: Medium (60%)' at the bottom sets expectations honestly — sending the message is not guaranteed to work, and the interface should never imply otherwise.

3.2 Screens 5-7: Trust, Explainability & the Broken-Promise Flow

5. Explainability Panel ("Why?")



6. Promise Logged



7. Promise Broken → Re-simulated

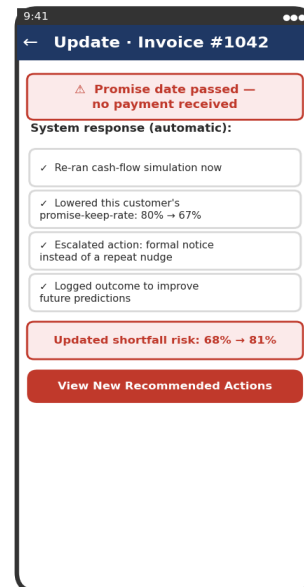


Figure 3. Screens 5-7 — Explainability panel, Promise logged, and the automatic re-simulation flow after a broken promise.

Screen 5 — Explainability Panel ("Why?")

- **Purpose:** answer 'why should I trust this score' on demand, without cluttering the primary screens — reachable via a 'Why?' link from any risk chip anywhere in the app.

- **Key element:** the three contributing-factor bars are a direct, simplified readout of the hazard model's feature contributions (Technical Design Doc, Section 4.2) — late-payment ratio, invoice-value deviation, recent trend. This is the single feature most likely to differentiate the product from a black-box competitor in a judge's eyes.
- **Key element:** the confidence footer ('Personalized — 18 prior invoices') ties this specific score back to the cold-start blending mechanism, making an abstract model concept concrete and visible.

Screen 6 — Promise Logged

- **Purpose:** let a user record that a customer verbally/by-message promised a payment date, without ever letting that promise silently become '100% certain' in the system.
- **Key element:** the amber callout stating the customer's historical promise-keep-rate (80% in this example) and explicitly stating the risk curve was adjusted, not zeroed out — this is the UI-level guarantee that the failure mode discussed earlier (a broken promise silently trusted as fact) cannot happen, because the interface itself shows the math never went to 100%.
- **Key element:** showing 'was 45% before the promise' next to the updated 70% gives the user a visible, honest before/after — reinforcing that this is a probability shift, not a guarantee.

Screen 7 — Promise Broken → Automatic Re-simulation

- **Purpose:** this is the screen that proves the system treats a broken promise as new evidence to learn from, not as a failure to hide.
- **Key element:** the checklist of automatic system responses (re-simulated, lowered promise-keep-rate, escalated the action tier, logged the outcome) makes the continuous-learning loop from the architecture visible and tangible to the user — turning an internal ML concept into a reassuring, transparent moment instead of a silent failure.
- **Key element:** the updated risk number (68% → 81%) shown with an explicit before/after arrow demonstrates the system reacting in real time, which is a strong, concrete answer to 'what happens when the input turns out to be wrong.'

4. Design System / Style Guide (for Figma tokens)

4.1 Color tokens

The palette is semantic first, decorative second — every color in the product means something specific about risk or confidence, and that mapping must stay consistent everywhere.

 Primary / Navy #1F3864	 Accent / Blue #2E74B5	 Safe / Green #2E7D32	 Caution / Amber #B8860B	 Risk / Red #9C2B2B
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Figure 4. Core semantic palette. Navy = brand/structure (app bars, headings). Blue = interactive/primary actions. Green = safe/confirmed/personalized-confidence. Amber = caution/cold-start/pending-promise. Red = high risk/shortfall.

- **Rule:** red is reserved exclusively for shortfall/high-risk states — never used decoratively, so its appearance always means 'pay attention.'
- **Rule:** amber specifically marks anything with reduced confidence — cold-start banners, pending promises, medium-impact actions — giving the user one consistent visual language for 'this is less certain.'
- **Accessibility:** verify the red/green/amber trio against a color-blind-safe check in Figma (use the Stark or Able plugin) and never rely on color alone — every risk chip must also carry a text label (HIGH/MEDIUM/AVAILABLE), as already shown in the wireframes.

4.2 Typography

Role	Suggested size (mobile)	Weight
Screen title (app bar)	16-18px	Bold
Section heading	14-15px	Bold
Body / card text	13-14px	Regular
Secondary / meta text (timestamps, sub-labels)	11-12px	Regular, muted grey
Large stat callouts (e.g. '68% chance short')	18-22px	Bold

- **Font recommendation:** a single, highly legible geometric sans-serif (Inter, or Noto Sans if regional-language support is needed early) — avoid condensed or decorative fonts given the target user's likely lower tolerance for visual friction.

4.3 Core reusable components to build first in Figma

Component	Variants needed	Used on screens
Probability Band Chart	Cold-start (wide band) / Personalized (narrow band) / Safe (green, no red threshold crossed)	Home, Insights (what-if)
Confidence Badge	Cold Start / Warming Up / Personalized — each with its own color and %	App bar on nearly every screen
Risk Score Chip	HIGH / MEDIUM / AVAILABLE / SAFE	Receivables list, Action cards, Explainability
Action Card	Chase / Delay / Finance — each with a different accent color and CTA verb	Actions tab, Shortfall detail
Message Draft Card	WhatsApp-styled / SMS-styled / Email-styled	Message draft & send
Factor Contribution Bar	Default / worsening-trend (accent color shifts)	Explainability panel
Promise Tracker Card	Pending / Kept / Broken (three distinct states, three	Receivables detail, Promise

Component	Variants needed	Used on screens
	colors)	flow
Alert Banner	Info (blue) / Caution (amber) / Risk (red) / Success (green)	Home, Actions, Promise flow
Bottom Nav Bar	4 tabs x active/inactive state each	Global

Build every component with Figma auto-layout and named variants from day one — this single decision is what makes a hackathon-speed prototype still look and behave consistently across 15+ screens instead of visibly diverging.

5. Interaction & State Rules

5.1 States every data-bearing screen must design for

State	What changes visually	Why it matters here specifically
Cold start	Wider probability bands, amber confidence badge, cohort-based copy ('businesses like yours')	Directly reflects the cohort-prior blending mechanism — must never look identical to a personalized, high-confidence screen
Warming up	Narrowing bands, badge shifts amber → blue, some cards show 'still learning' microcopy	Makes the continuous-learning loop visible as it happens, not just at the two extremes
Personalized / high confidence	Narrow bands, green/blue badge, no hedging microcopy needed	Reward state — gives the user a visible reason to keep using the product
Loading / re-simulating	Skeleton loaders on charts, a small 're-calculating...' inline label rather than a full-screen spinner	Re-simulation happens live after events like a broken promise; it should feel responsive, not disruptive
Empty (no invoices yet)	Friendly illustration + a direct CTA to add the first invoice	Cold-start users may have zero data initially; never show a blank chart with no explanation
Error / offline	Cached last-known state clearly labeled 'last updated Xh ago', no fabricated live numbers	Given patchy connectivity in the target segment, never silently show stale data as if it were current

5.2 Micro-interactions worth prototyping in Figma (not just static frames)

- **Pull-to-refresh on Home** triggers a visible 're-simulating your cash flow...' state — make this a real prototyped transition, since it's the clearest way to demo that Layer 2 is a live simulation, not a cached report.
- **Tapping any risk chip anywhere** always opens the same Explainability panel component as a bottom sheet — consistent entry point trains the user that 'why' is always one tap away.
- **Swipe-to-act on the Receivables list** (swipe left on an invoice row to reveal 'Chase' shortcut) mirrors familiar messaging-app gestures for this WhatsApp-native audience.
- **The one-tap Send button** should have a distinct sent-confirmation animation (checkmark + brief toast) — this is the emotional payoff moment of the whole flow and deserves deliberate motion design, not an instant silent state change.

6. Figma File Organization Plan

A concrete structure to set up before drawing a single screen — this is what keeps a hackathon-speed Figma file navigable for both the team and the judges.

6.1 Pages (top-level tabs inside the Figma file)

Page name	Contents
00 — Cover & Overview	Project name, one-line pitch, this document's screen map thumbnail for quick orientation
01 — Design Tokens	Color styles, text styles, spacing/grid guide — defined as real Figma Styles/Variables, not just a reference image
02 — Components	Every component from Section 4.3, built with auto-layout and all variants, kept isolated from screen frames
03 — Flows (low-fi)	Quick box-and-arrow user flow diagrams (can literally reuse the structure of Figures 1-3 in this doc) — agree on flow before polishing visuals
04 — Screens (Mobile, Hi-Fi)	All final screens, organized in horizontal rows per flow (Onboarding row, Core Loop row, Trust/Explainability row)
05 — Prototype	A duplicated, cleaned set of frames wired with prototype connections — keep this separate from the working design page so demo links never break while design is still being edited

6.2 Naming conventions

- **Screens:** "[Flow]/[Number]-[ScreenName]" e.g. Core Loop/02-Home-Dashboard, Trust/07-Promise-Broken.
- **Components:** "Component/[Name]/[Variant]" e.g. Component/RiskChip/High, Component/ConfidenceBadge/ColdStart — this naming is what makes Figma's variant switcher usable instead of a flat unsorted list.
- **Color styles:** semantic names, not hex-based — "Risk/High", "Confidence/Personalized", never "Red-1" — so switching a shade later doesn't require renaming anything referencing it.

6.3 Building the clickable prototype (for the judge demo)

Wire exactly one primary path start-to-finish first, before branching out to secondary flows — a single flawless path beats five half-connected ones in a live demo.

Recommended demo path (under 2 minutes, matches the wireframes in this doc)

- 1. Home Dashboard → tap the red shortfall banner
- 2. Shortfall Detail & Actions → tap 'Send Nudge' on the top action card
- 3. Message Draft & Send → tap 'Send Reminder' → confirmation animation
- 4. Back to Home → tap the risk chip's 'Why?' link → Explainability Panel opens as a bottom sheet
- 5. (Optional stretch branch) → Promise Logged → Promise Broken → Re-simulation screen, to demonstrate the system's response to changing evidence live if there's time in the pitch

This exact path was chosen because it demonstrates prediction, action, automation, and explainability in under five taps — the four things Section 8 of the Technical Design Document identifies as what separates this from a generic predict-and-alert pitch.

7. Accessibility & Localization Notes

- Minimum touch target 44x44px on all buttons and chips — the target user may be tapping one-handed, in a shop, mid-transaction.
- Design a regional-language toggle into Settings from the start (even if only English + Hindi ship for the demo) — retrofitting localization later means re-touching every text layer.
- Keep sentence-level copy short and concrete (as shown in the wireframes) — avoid financial jargon like 'accounts receivable' in user-facing copy; use 'money customers owe you' instead.
- Verify all semantic risk colors pass a color-blindness check, and never let color be the only signal (text labels on every chip, as already specified).
- Design the offline/error state (Section 5.1) early — do not treat it as a low-priority edge case, given the target segment's likely connectivity constraints.